Math 248 Project: Professor Evaluations

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Professor Evaluations, Beauty Ratings, and Gender

Student evaluations are often used to measure teaching quality, but they may also reflect other things about how students perceive professors. One possible factor is physical appearance. The `evals` dataset from the `moderndive` package includes end-of-semester student evaluations from the University of Texas at Austin. The data include 463 courses taught by 94 professors. Each row represents one course and includes information about the course, the professor, the professor's teaching evaluation score, and the professor's average beauty rating.

The beauty ratings were based on ratings from six students. These data were used in a published study by Hamermesh and Parker about whether professors' physical appearance is associated with higher teaching evaluations.

In this project, we are interested in whether the relationship between beauty rating and teaching evaluation score differs by professor gender.

Our research question is:

Does the relationship between a professor's average beauty rating and their teaching evaluation score differ by professor gender?

We predict that beauty rating and evaluation score will have a positive association. We also think this relationship may differ by gender, meaning beauty rating may be more strongly related to evaluation scores for one gender group than the other.

```
data(evals)

evals <- evals %>%
  mutate(gender = as.factor(gender))

head(evals)
```

```
# A tibble: 6 x 14
     ID prof_ID score   age bty_avg gender ethnicity   language rank  pic_outfit
  <int>   <int> <dbl> <int>   <dbl> <fct>  <fct>       <fct>    <fct> <fct>
1     1       1   4.7    36       5 female minority    english  tenu~ not formal
2     2       1   4.1    36       5 female minority    english  tenu~ not formal
3     3       1   3.9    36       5 female minority    english  tenu~ not formal
4     4       1   4.8    36       5 female minority    english  tenu~ not formal
5     5       2   4.6    59       3 male   not minori~ english  tenu~ not formal
6     6       2   4.3    59       3 male   not minori~ english  tenu~ not formal
# i 4 more variables: pic_color <fct>, cls_did_eval <int>, cls_students <int>,
#   cls_level <fct>
```

#Analysis

In this analysis, we focus on `score` as the response variable and `bty_avg` and `gender` as explanatory variables.

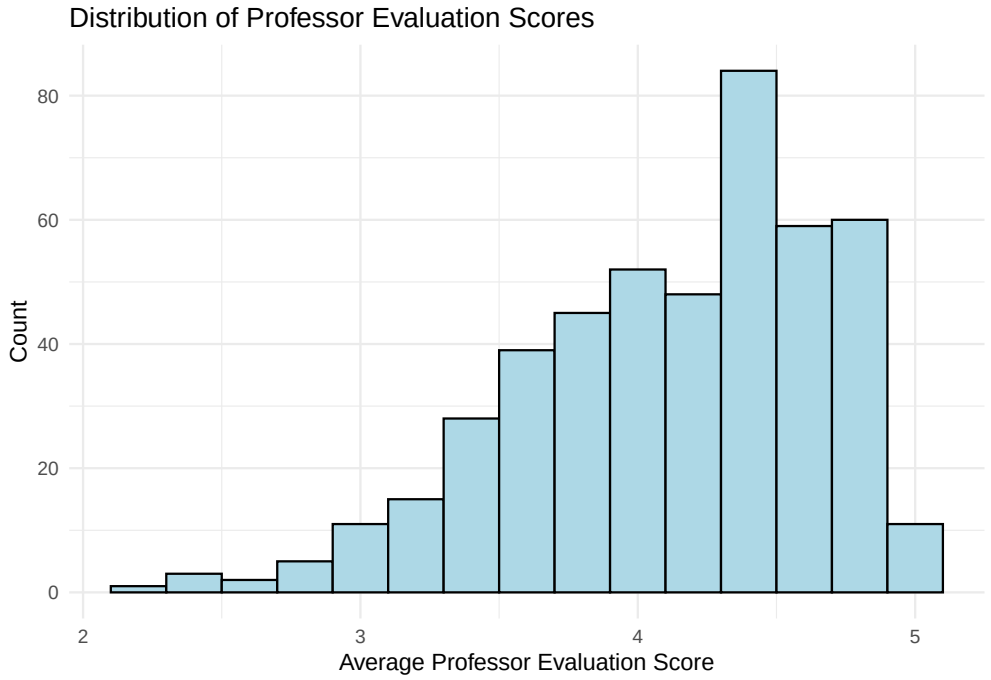
The response variable is `score`, the average professor teaching evaluation score. It is quantitative and measured on a scale from 1 to 5, where 1 means very unsatisfactory and 5 means excellent.

The quantitative explanatory variable is `bty_avg`, the professor's average beauty rating. It is measured on a scale from 1 to 10.

The categorical explanatory variable is `gender`. It has two levels, female and male.

We first look at the distribution of the response variable.

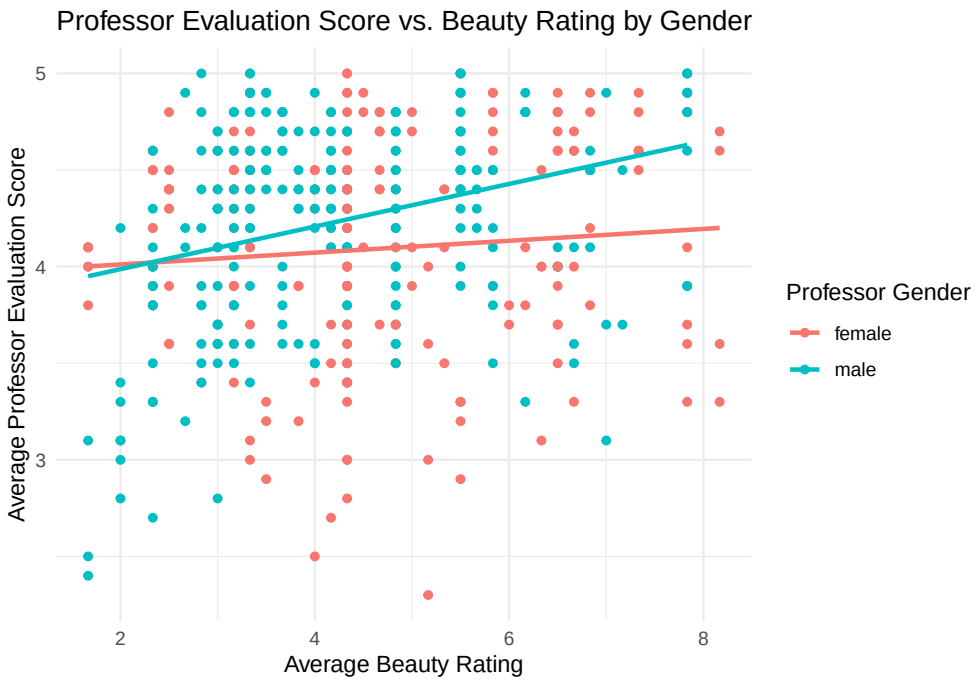
```
ggplot(evals, aes(x = score)) +
  geom_histogram(binwidth = 0.2, color = "black", fill = "lightblue") +
  labs(title = "Distribution of Professor Evaluation Scores",
       x = "Average Professor Evaluation Score",
       y = "Count") +
  theme_minimal()
```



The histogram is slightly left-skewed. The center is around 4.0, and the scores range from about 2.3 to 5. There are a few lower scores, but nothing looks very unusual.

We also create a visualization to see how beauty rating and gender affect evaluation score. A scatterplot of `score` by `bty_avg`, with color indicating `gender`.

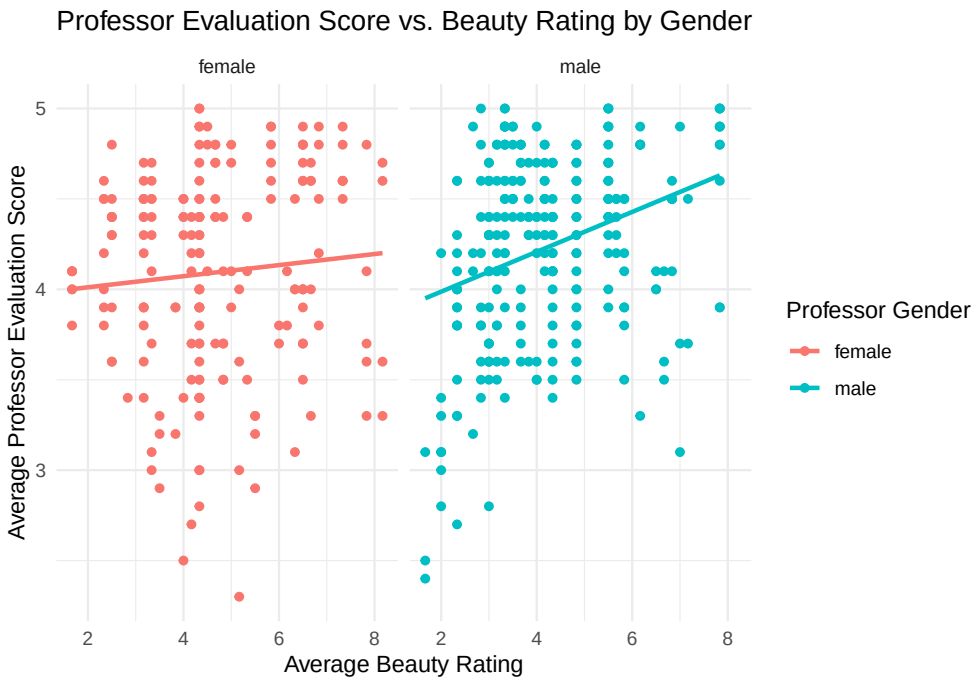
```
ggplot(evals, aes(x = bty_avg, y = score, color = gender)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Professor Evaluation Score vs. Beauty Rating by Gender",
    x = "Average Beauty Rating",
    y = "Average Professor Evaluation Score",
    color = "Professor Gender"
  ) +
  theme_minimal()
```



The plot shows a positive association between beauty rating and evaluation score. However, the lines do not look parallel. The male line looks steeper than the female line. This suggests that the relationship between beauty rating and evaluation score may be different for male and female professors.

We also separate the plots by gender.

```
ggplot(evals, aes(x = bty_avg, y = score, color = gender)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ gender) +
  labs(
    title = "Professor Evaluation Score vs. Beauty Rating by Gender",
    x = "Average Beauty Rating",
    y = "Average Professor Evaluation Score",
    color = "Professor Gender"
  ) +
  theme_minimal()
```



The separated plots also suggest that the slopes may be different. For male professors, the relationship between beauty rating and evaluation score appears stronger. For female professors, the relationship looks positive, but much weaker.

In this case, an interaction would mean that the relationship between beauty rating and evaluation score changes based on professor gender.

Because the plots suggest different slopes, we fit a model that includes an interaction between `bty_avg` and `gender`.

The theoretical model is:

$$Score_{gender} = Intercept_{gender} + Linear\ Slope\ Coefficient_{gender} \times bty_avg + Error$$

This is the separate-slopes model that assumes different intercepts and different slopes for each gender.

The model conditions are:

- Linearity
- Independent observations
- Normally distributed residuals
- Equal SD/variances

Statistical Model and Results

Let's fit a model which includes an interaction term between the gender variable and the beauty rating variable.

```
interaction_model <- lm(score ~ bty_avg * gender, data = evals)
summary(interaction_model)
```

```
Call:
lm(formula = score ~ bty_avg * gender, data = evals)
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-1.8084 -0.3828  0.0903  0.4037  0.9211

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         3.95006    0.11800  33.475   <2e-16 ***
bty_avg             0.03064    0.02400   1.277   0.2024
gendermale         -0.18351    0.15349  -1.196   0.2325
bty_avg:gendermale  0.07962    0.03247   2.452   0.0146 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5258 on 459 degrees of freedom
Multiple R-squared:  0.07129,   Adjusted R-squared:  0.06522
F-statistic: 11.74 on 3 and 459 DF,  p-value: 1.997e-07
```

```
tidy(interaction_model)
```

```
# A tibble: 4 x 5
  term               estimate std.error statistic   p.value
  <chr>                 <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)          3.95      0.118      33.5  2.92e-125
2 bty_avg              0.0306    0.0240      1.28 2.02e-  1
3 gendermale          -0.184     0.153      -1.20 2.32e-  1
4 bty_avg:gendermale   0.0796    0.0325      2.45 1.46e-  2
```

We fitted a separate-slopes (interaction) model to determine if the effect of beauty on evaluation scores differs by gender.

While the individual predictors of bty_avg (p=0.202) and gender (p=0.233) were not statistically significant on their own in this interaction model, the interaction term (bty_avg:gendermale) was statistically significant (p=0.0146). This significance indicates that the slope for male professors is statistically different from the slope for female professors.

The model explains approximately 7.1% of the variation in evaluation scores (R^2=0.071) while this interaction is statistically significant, the model's R^2 of 0.071 indicates that beauty and gender only account for about 7.1% of the variation in scores. The remaining 92.9% of the variation is likely explained by factors not included in this specific model, such as teaching style, course difficulty, or student engagement.

The non-significant `gendermale` term means that when beauty rating is 0, the predicted difference between male and female professors is not statistically significant. Since a beauty rating of 0 is not meaningful in this dataset, this term is not very useful to interpret on its own.

Based on the interaction model, the estimated regression equation is:

$$\widehat{score} = 3.9501 + 0.0306(bty_avg) - 0.1835(male) + 0.0796(bty_avg \times male)$$

Using female professors as the reference group, female professors have `male = 0`.

So the equation for female professors is:

$$\widehat{score}_{female} = 3.9501 + 0.0306(bty_avg)$$

For male professors, `male = 1`.

So the equation for male professors is:

$$\widehat{score}_{male} = (3.9501 - 0.1835) + (0.0306 + 0.0796)(bty_avg)$$

This simplifies to:

$$\widehat{score}_{male} = 3.7666 + 0.1102(bty_avg)$$

For female professors, a point increase in beauty rating is associated with about a 0.031 increase in predicted evaluation score. This is a very weak positive association, and the p-value for the female slope is 0.202, so this relationship is not statistically significant by itself.

For male professors, a point increase in beauty rating is associated with about a 0.110 increase in predicted evaluation score. This slope is much steeper than the female slope, which suggests that beauty rating is more strongly related to evaluation score for male professors.

The interaction coefficient is about 0.0796. This means that the slope for male professors is about 0.080 points larger than the slope for female professors. Since the interaction term is statistically significant, the relationship between a professor's average beauty rating and their teaching evaluation score does differ by professor gender at the University of Texas at Austin.

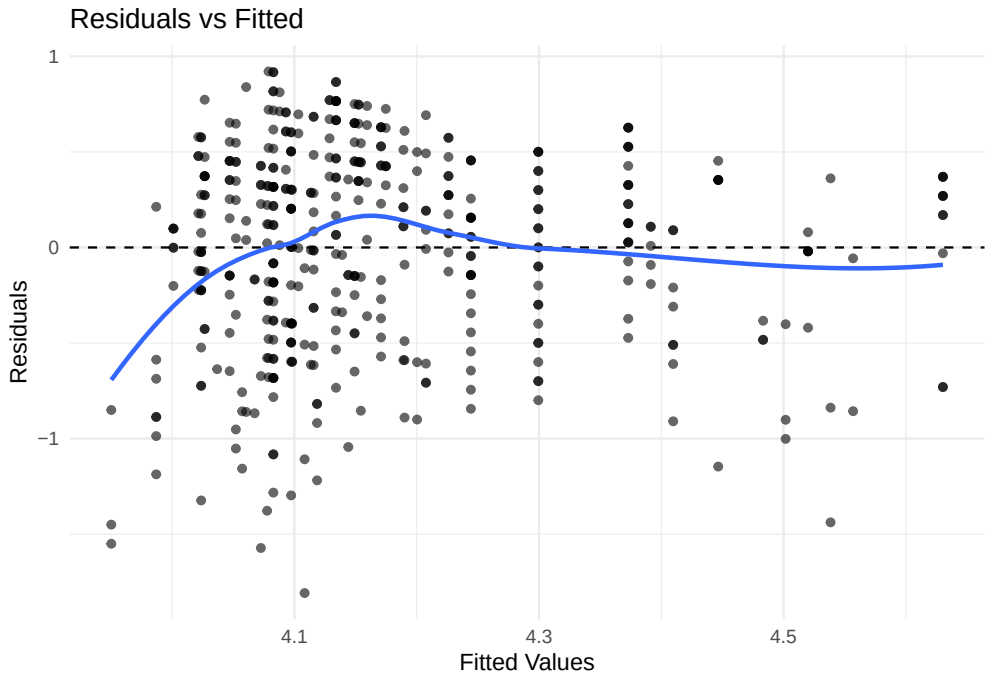
Overall, beauty rating has a small positive association with evaluation score for female professors and a stronger positive association for male professors. This suggests that the "beauty premium" in teaching evaluations appears more clearly for male professors than for female professors in this dataset.

Model Conditions

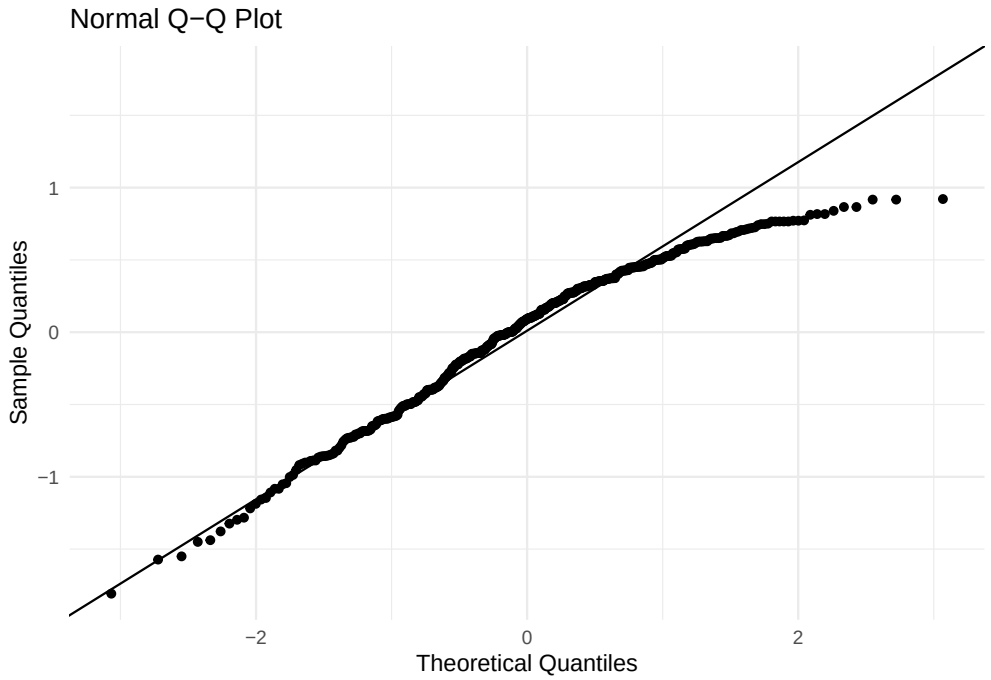
To assess the fit of this model, we looked at residual plots.

```
model_diag <- fortify(interaction_model)

ggplot(model_diag, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(se = FALSE) +
  labs(title = "Residuals vs Fitted",
       x = "Fitted Values",
       y = "Residuals") +
  theme_minimal()
```




```
ggplot(model_diag, aes(sample = .resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Normal Q-Q Plot",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```



The linearity condition seems reasonable because the residuals are scattered around 0 with no clear strong curve. The equal variance condition seems mostly reasonable because the spread is fairly similar across fitted

values. The normality condition seems reasonable, although the Q-Q plot has some deviations at the tails. The independence condition is reasonable if each course evaluation was recorded independently, but some professors appear more than once in the dataset, so this is a limitation.

Conclusions

The relationship between beauty rating and professor evaluation score differs by professor gender. For female professors, beauty rating has a small positive association with evaluation score. For male professors, beauty rating has a stronger positive association with evaluation score.

The interaction model suggests that beauty rating is more strongly related to evaluation scores for male professors than for female professors. However, the model only explains a small amount of the variation in evaluation scores, so beauty rating and gender are not the only factors related to teaching evaluations.

These results should only be generalized cautiously to similar university course evaluation settings, especially because the data come from courses taught at one university. Since this is an observational study, we cannot say that beauty rating or gender causes higher teaching evaluation scores.